

Particulate Matter Monitoring & Health Effects

Issue 2026-08-31 | Entry window 31 August 2026 (single day) | 16 records in scope, 31 screened out

16

RECORDS IN SCOPE
(31 SCREENED OUT)

7

SUBTOPIC CLUSTERS
OF TEN OCCUPIED

5

EXPOSURE ASSES-
MENT
& MODELLING

2

EFFECT ESTIMATES
(ONE SOURCE)

10

RECORDS WITH NO
HEALTH ENDPOINT

Signal of the day

Two *ES&T* papers published on the same day argue, from opposite directions, that $\text{PM}_{2.5}$ mass is the wrong control variable — one on toxicity weighting, one on the exposure surfaces that feed it.

Song et al. make the policy case: age-standardised $\text{PM}_{2.5}$ -attributable death rates have diverged worldwide over the past decade even as mass concentrations fell, which is *prima facie* evidence against the uniform-toxicity assumption written into every current standard. Weighting exposure by source-specific toxicity — incomplete combustion, especially residential solid fuel and wildfire, carrying the highest toxicity per unit mass — puts **24.0% of the global toxicity-adjusted exposure burden on the 8.2% of the population that is low-income**. Li Q et al. supply the other half of the machinery for China: a 1 km daily $\text{PM}_{2.5}$ surface for 2013–2023 (cross-validated R^2 0.88 spatially, 0.82 temporally) fused with meta-analysed short-term effect estimates into a pollutant-specific AQHI. National mean $\text{PM}_{2.5}$ fell **55.06 → 26.50 $\mu\text{g m}^{-3}$** (2013 → 2022) then rebounded to 27.46 in 2023; per 10 $\mu\text{g m}^{-3}$ over three days the associated increases were 0.42% all-cause, 0.61% cardiovascular and 0.84% respiratory mortality. Note what neither paper delivers: a *measurement* path to source-resolved toxicity. Both weight after the fact, from inventories and meta-analysis. For distributed low-cost sensing this is the open brief — speciation or source discrimination at network cost, not another mass channel.

What changed today

- **An unusually modelling-heavy day.** *Exposure assessment & modelling* is the largest cluster (5/16) and *Modelling / inventory* the largest architecture (5/16); ten of sixteen records carry no human health endpoint at all. Only one record (Wang L et al.) reports an effect estimate with a confidence interval, so today's forest plot has a single source and is not a synthesis.
- **HEI's multinational exposure-assessment initiative reports out.** Boogaard et al. synthesise the whole programme: exposure surfaces that add daily mobility and outdoor–indoor infiltration correlate >0.7 with conventional residential estimates, and the resulting health effect estimates differ in *magnitude but not direction*. That is a genuinely useful null for anyone deciding whether a mobility-aware exposure model is worth its cost — and it is the strongest argument in today's corpus for spending effort on validation data rather than on more model structure.
- **Sulfate chemistry got two independent corrections, both pointing the same way.** Liu T et al. show aerosol solute strength drives Mn(II)-catalysed sulfate formation **more than two orders of magnitude faster** than dilute-cloudwater parameterisations predict, raising modelled haze-episode sulfate by $>50\%$; Guo et al. show that ignoring kinetic isotope effects in source apportionment overstates coal combustion by 10% and understates traffic by 8% in summer. Both say the transition-metal pathway has been systematically under-credited.
- **A cohort result with a real interaction term.** Wang L et al. (CHARLS, $n=6,457$, 710 strokes) report **HR 1.321 (1.222–1.428) per 1 SD annual $\text{PM}_{2.5}$** , a significant multiplicative interaction with small dense LDL-C ($p=0.037$), and **HR 1.713 (1.335–2.198)** for the dual-high stratum. The lipid fraction, not total LDL-C, is what carries the interaction (AUC 0.578 vs. 0.534).
- **Sub-Saharan Africa is still absent; North Africa is not.** Saad et al. report a two-campaign, 24-site port–urban survey at Alexandria — the first African record since 29 August. Particulate levels were low across the study area; the port signal is in CO and NH_3 , not PM, which the authors state plainly.
- **Two records ship on metadata only.** Both indoor-air papers (10.1016/j.buildenv.2026.115193, 10.1016/j.indenv.2026.100191) deposited no abstract to Crossref and are not yet in PubMed, Europe PMC or Semantic Scholar. They are listed and counted but not summarised, and no numbers are attributed to them.

Corpus analytics

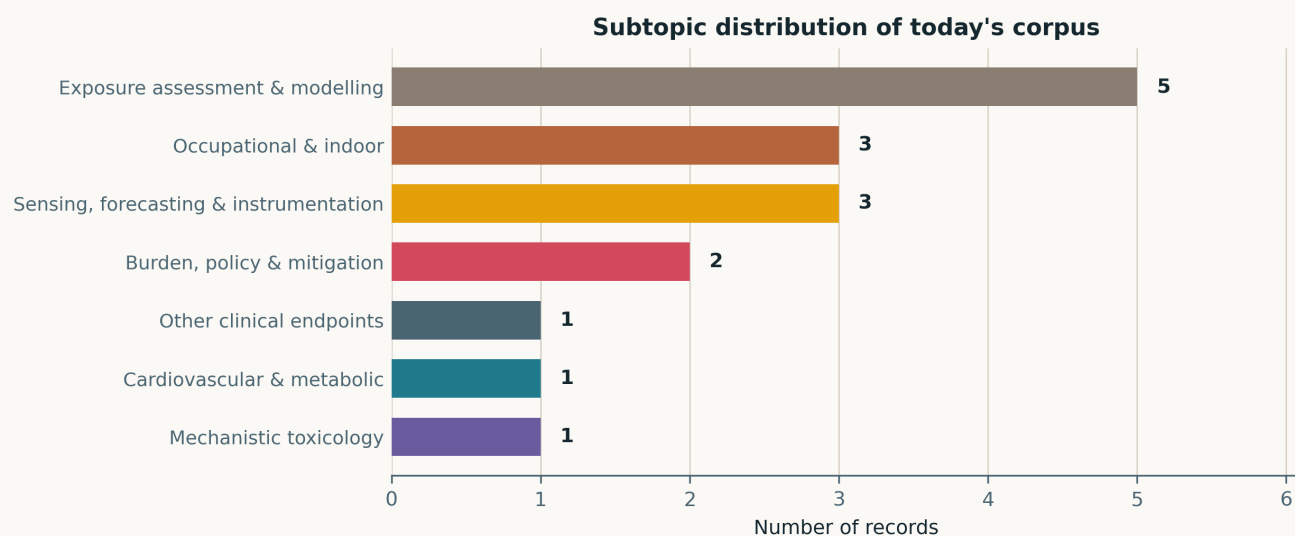


Fig. Subtopic assignment of the 16 in-scope records. Seven of the ten clusters are occupied; *Neuro / mental health*, *Respiratory & allergic* and *Reproductive & developmental* are all empty today — the second issue in which all three are simultaneously vacant, after 5 August. *Exposure assessment & modelling* leads with 5, which is the day's defining feature rather than an artefact of labelling: three of those five (Guo, Chi, Hernández Pardo) are atmospheric-process papers with no health endpoint at all.

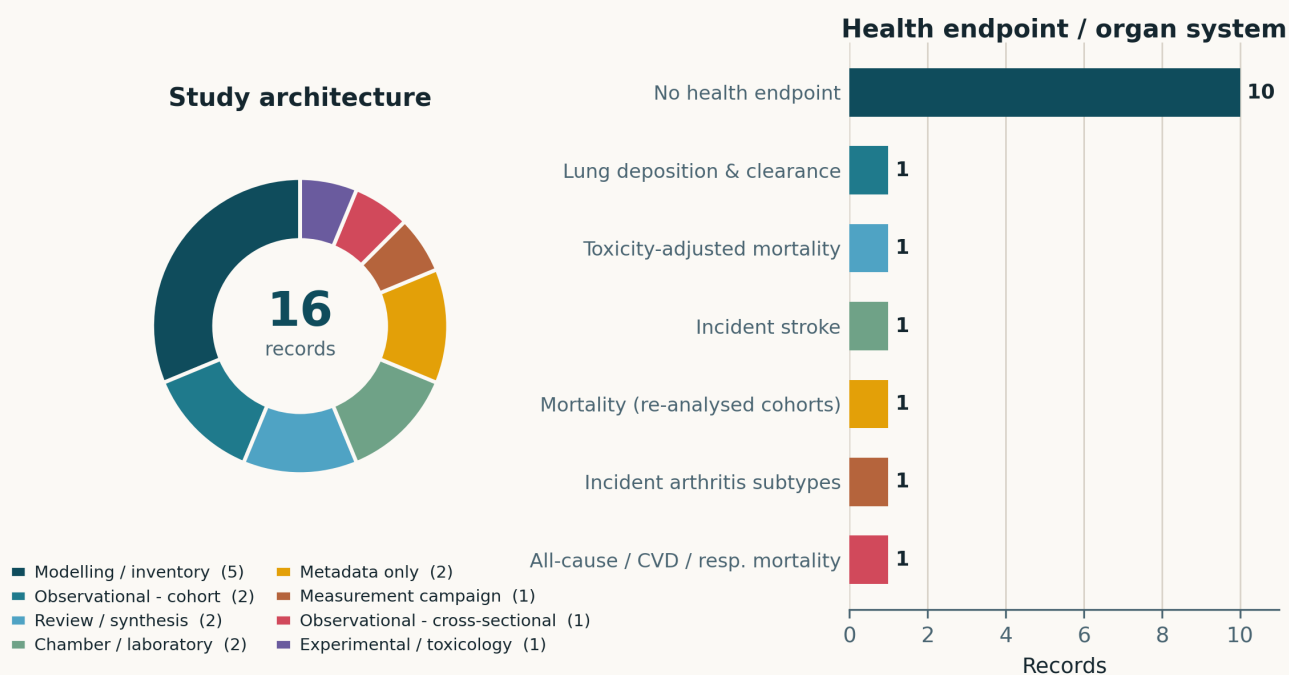


Fig. Left — study architecture. Modelling and inventory work is the largest single group (5/16), with chamber/laboratory (2), cohort (2) and review (2) behind it; the two *Metadata only* entries are the abstract-free indoor-air deposits and should be read as an absence of information, not as a design. Right — health endpoint by organ system. **Ten of sixteen records (62%) carry no human health endpoint** — among the highest shares this month, though short of 20 August's 72%; the remaining six split across musculoskeletal, cardiovascular, respiratory-dosimetry and all-cause mortality.

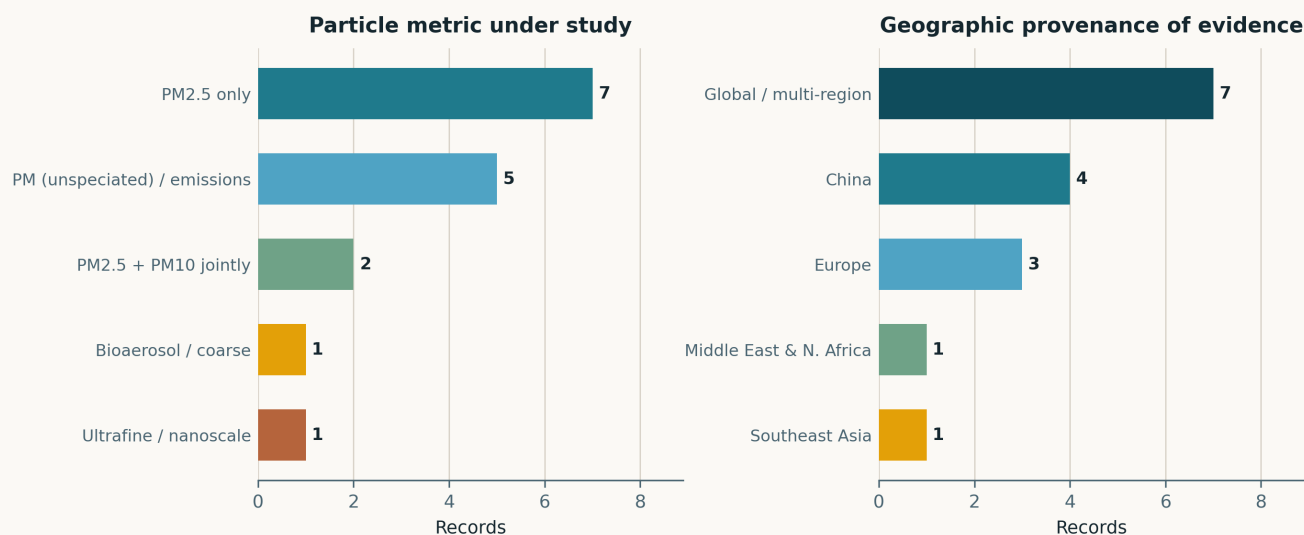


Fig. Left — particle metric. PM_{2.5} alone is the exposure in 7 records and appears jointly with PM₁₀ in 2 more; one record (Garcia Dominguez et al.) is explicitly ultrafine, and one is bioaerosol/coarse. Right — geographic provenance. *Global / multi-region* is the largest bucket at 7, which here reflects the modelling weight of the day rather than genuine geographic spread — five of those seven are simulations or global syntheses with no field site. China contributes 4, Europe 3 (two UK, one Sweden), Southeast Asia 1 (Malaysia) and Middle East & North Africa 1 (Egypt). *Sub-Saharan Africa, South Asia and Latin America are all absent.*

Life-course windows addressed today (records may span several stages)

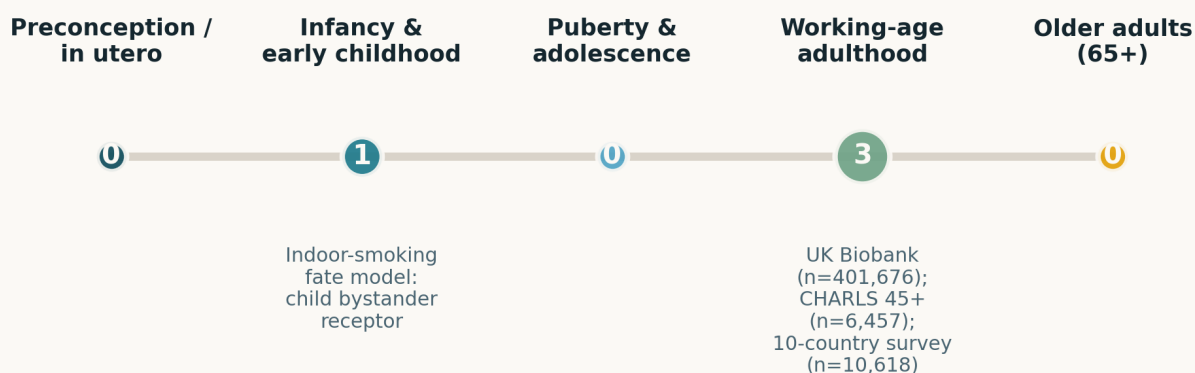


Fig. Life-course windows addressed. Counts are non-exclusive and are deliberately sparse today: only four records study identifiable human populations. **No record in this issue reports an age-stratified estimate**, so the 65+ cell is scored zero even though both cohorts (UK Biobank, median 13 y follow-up from baseline age 40–69; CHARLS, 9 y from age 45+) certainly contain large older strata. Scoring those as 65+ records would misdescribe what the papers report.

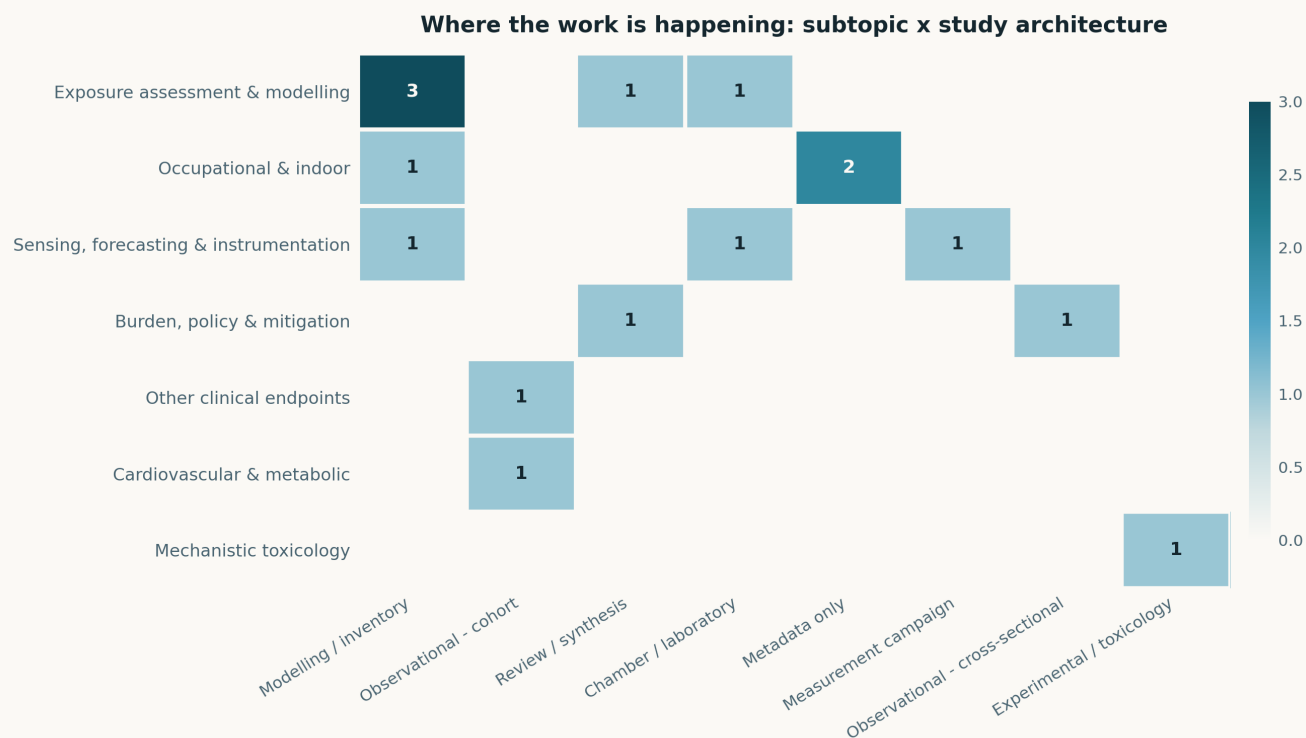


Fig. Subtopic × study-architecture cross-tabulation. The structure to note is the near-total separation of the modelling column from any health-endpoint row: *Exposure assessment & modelling* and *Sensing, forecasting & instrumentation* together hold 8 of 16 records and contribute *zero* effect estimates. The two cohort records sit alone in *Cardiovascular & metabolic* and *Other clinical endpoints*. This is the instrumentation–epidemiology siloing this watch has flagged repeatedly, and today it is close to complete.

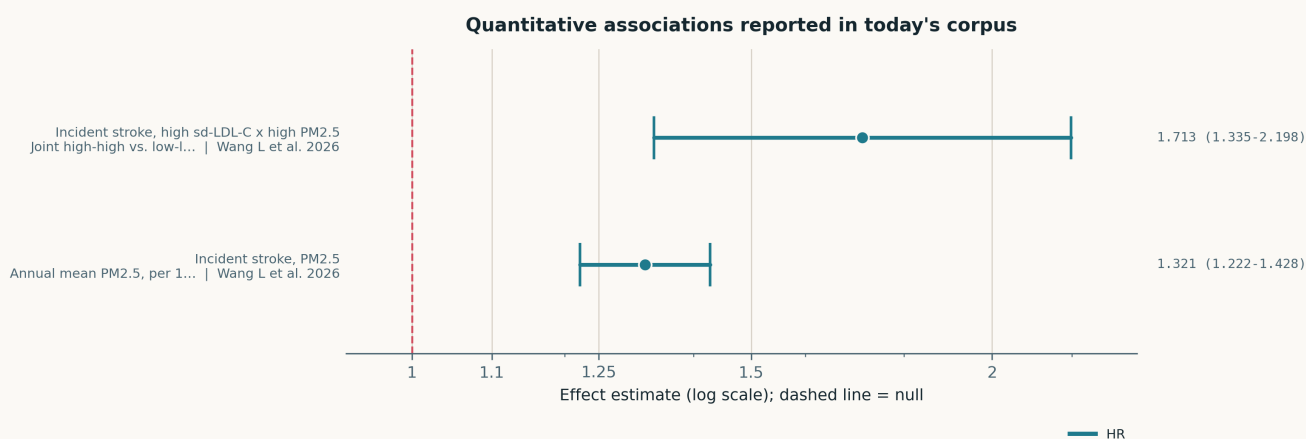


Fig. Both estimates come from a single paper (Wang L et al., CHARLS) and this figure is therefore a display of that study's two contrasts, not a synthesis across the corpus. The per-1 SD hazard ratio and the dual-high-stratum hazard ratio are not independent — the second is a subset contrast of the same cohort — so the apparent agreement between them carries no replication weight. No other record in this issue reported an effect estimate with a confidence interval.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Exposure assessment & modelling

5 records

Li Q et al. 2026 Environ Sci Technol

Health risk of fine PM across China with a pollutant-specific index, 2013–2023

A 1 km daily PM_{2.5} surface for 2013–2023 built by machine learning (cross-validated R^2 **0.88 spatially, 0.82 temporally**), fused with meta-analysed short-term effect estimates into an AQHI_{PM2.5}. National mean fell **55.06 → 26.50 $\mu\text{g m}^{-3}$** (2013→2022) then rebounded to 27.46 in 2023. Per 10 $\mu\text{g m}^{-3}$ over 3 d: **+0.42% all-cause, +0.61% cardiovascular, +0.84% respiratory mortality**. Northern and southwestern regions retain >80 medium/high-risk days a year; eliminating them would avert 6,985 (6,419–7,588) deaths, 7.42% of the attributable burden. The weak point is the meta-analysed exposure — response step, which imports between-study heterogeneity into a spatially resolved index and then reports it at 1 km as if the precision were the model's. For sensing work: the AQHI construction is directly portable to a dense network, but only if the network can supply the exposure surface it assumes.

doi: 10.1021/acs.est.6c00144

Boogaard et al. 2026 Curr Environ Health Rep

Advancing long-term outdoor air pollution exposure assessment for health studies

Overview of the completed HEI multinational initiative on novel long-term exposure assessment across North America and Europe. Exposure predictions from models that account for daily mobility patterns and outdoor–indoor infiltration correlated **>0.7** with conventional residential estimates, and the health effect estimates they produced were similar — differing in magnitude but *not direction*. The paper is honest that this is a comparison among models, not a validation against measurement: representative validation data for long-term exposure simply does not exist, which is the initiative's own headline gap. Read as the strongest current argument for spending resources on validation campaigns rather than additional model structure, and as a caution against over-reading small differences between LUR variants in cohort re-analyses.

doi: 10.1007/s40572-026-00560-8

Guo et al. 2026b Atmos Chem Phys

Kinetic isotope fractionation framework for sulfate source apportionment

Seasonal PM_{2.5} observations in Nanjing combined with Bayesian isotope mixing and process-specific Rayleigh fractionation. Conventional isotope models assume complete SO₂ oxidation and therefore discard the kinetic isotope effects generated during incomplete processing; correcting for them shows transition-metal-ion-catalysed and NO₂-mediated pathways dominate secondary sulfate, and moves apportionment by **–10% coal combustion, +8% traffic** during summer photochemical episodes. Single-city, single-year, and the Rayleigh treatment assumes a closed reservoir, which is a strong assumption for an urban boundary layer. The direction of the correction nevertheless agrees with Liu T et al. in the same issue, from wholly independent evidence.

doi: 10.5194/acp-26-12231-2026

Chi et al. 2026 Atmos Chem Phys

Heterogeneous nitrosation of amines by N₂O₄: a missing source of particulate nitrosamines

Born–Oppenheimer molecular dynamics plus metadynamics on methylamine and dimethylamine reacting with N₂O₄ at the air–water interface. Two competing routes: a *barrierless* direct N-nitrosation yielding nitrosamine cations and NO₃[–], and an amine-mediated N₂O₄ hydrolysis generating interfacial HONO in **2–16 ps**, which then nitrosates over a 7.65 kcal mol^{–1} barrier at 300 K. Purely computational, two amines, idealised interface — there is no measurement here and no yield that a chemical transport model can use directly. Its value is as a mechanism for particulate carcinogen formation that current CTMs omit entirely.

doi: 10.5194/acp-26-12261-2026

Hernández Pardo et al. 2026 Atmos Chem Phys

Downward transport of tropical upper-tropospheric aerosols

Multi-year global EMAC simulations releasing 19 idealised tracers in the tropical upper troposphere under resolved advection, parameterised convection, turbulent mixing and deposition. Age of air at 500 hPa is $\approx$ 45 d for tropics-wide sources but >250 d for regional continental ones — transport diagnostics are dominated by source extent, not by transport physics. Advection dominates; injection height matters more than particle size in the 20–100 nm range. Tracers are idealised, carry no mass or composition, and there is no surface validation, so this constrains *where* upper-tropospheric particles arrive, not *how much* they contribute to surface PM. Carried at tier C for that reason.

doi: 10.5194/acp-26-12275-2026

Sensing, forecasting & instrumentation

3 records

Liu T et al. 2026 Environ Sci Technol

Aerosol solute strength drives TMI-catalysed sulfate aerosol formation

Aerosol flow-reactor experiments on atmospherically relevant aqueous aerosols, decoupling solute strength, interfacial chemistry and Fe(III)–Mn(II) interaction. Sulfate formation from Mn(II) chemistry runs **>2 orders of magnitude faster** than models assuming dilute cloudwater; substantial insoluble Fe(III) suppresses the rate below an ionic-strength threshold of **16 mol kg^{–1}**. Implemented in an atmospheric model, this raises sulfate abundance during polluted haze episodes by **>50%**. The flow reactor cannot reproduce real aerosol organic coatings or mixing state, so the transfer to ambient particles is an extrapolation — but this is a rare case of a laboratory rate measurement closing a known model–measurement gap rather than opening a new one.

doi: 10.1021/acs.est.6c09521

Yohanna & Lim 2026 Chemosphere

Regional and seasonal PM_{2.5} and ozone variation across Malaysia, XGBoost + SHAP

Daily DOE network observations 2018–2023 for five Malaysian regions, augmented with ERA5 boundary-layer height, solar radiation and cloud cover; XGBoost against a multiple linear regression baseline, interpreted with SHAP. PM_{2.5} R^2 =**0.576**, RMSE **4.87 $\mu\text{g m}^{-3}$** ; O₃ R^2 =0.543, RMSE 3.4 ppb — rising to R^2 =0.85 only when persistence is included, which is the number to be sceptical of, since a persistence term will carry most of the skill in any daily-mean series. Kruskal–Wallis confirms regional (χ^2 =3239.5) and seasonal (χ^2 =477.8) structure. SHAP attributes PM variability to combustion tracers and meteorology, ozone chiefly to humidity and temperature. Useful as a tropical-monsoon reference for anyone tuning a regional model; not a calibration result.

doi: 10.1016/j.chemosphere.2026.145074

Saad et al. 2026 Environ Monit Assess*Short-term air pollutant monitoring at the Alexandria port–urban exposure zone*

Two monitoring campaigns, May–October 2022, across 11 in-port and 13 surrounding urban sites, using a portable multi-gas monitor and particulate sensors, with hourly meteorology from Ras El-Tin (campaign-mean wind $3.0\text{--}7.2\text{ m s}^{-1}$, northerly to north-westerly). Fuel and bulk-fertiliser berths recorded significantly higher CO and NH₃ than urban sites; SO₂ and traffic pollutants outside the port dominate the urban background. **Particulate levels were low across the whole study area** — the authors say so, and the paper is correspondingly framed as screening-level spatial evidence, not exposure assessment. No sensor model, no reference co-location and no uncertainty budget are reported, which caps what the PM numbers can support. Carried mainly because African monitoring records are rare in this corpus.

doi: 10.1007/s10661-026-15828-7

Occupational & indoor exposure**3 records****Illa et al. 2026** Environ Sci Technol*Exposure pathways when psychoactive drugs are smoked indoors*

Mechanistic indoor chemical-fate and exposure model across seven compounds spanning orders of magnitude in volatility and hydrophobicity, with scenarios varying smoker retention rate and smoking frequency, resolving first-, second- and third-hand pathways and inhalation, mouth-ingestion and dermal routes. Dominant pathway is strongly chemical-specific: volatile compounds (nicotine) go by inhalation; intermediate hydrophobicity by dermal absorption; involatile hydrophobic compounds (Δ^9 -THC) receive meaningful contributions from all three. For non-users the SHS/THS balance is set jointly by partitioning and by the ratio of active-smoking to post-smoking time. Entirely modelled — no measurements, no validation — so the useful output is the ranking of pathways, not the doses.

doi: 10.1021/acs.est.6c02953

Jin et al. 2026b Build Environ*Air change rate and air purifier type for bioaerosol control during dental ultrasonic scaling*

Metadata only. The Crossref deposit carries no abstract and the record is not yet in PubMed, Europe PMC or Semantic Scholar. Listed and counted for completeness; no finding, effect size or method detail is attributed to it, and it is on the abstract retry list.

doi: 10.1016/j.buildenv.2026.115193

Wood et al. 2026b Indoor Environments*Influence of ventilation rate on the potential for indoor particulate matter exposure*

Metadata only. Same situation — no abstract in the Crossref deposit and no other index carries it yet. Flagged here because the title is squarely in scope for indoor PM sensing; it is on the abstract retry list and will be summarised if the abstract appears.

doi: 10.1016/j.indenv.2026.100191

Burden, policy & mitigation**2 records****Song et al. 2026** Environ Sci Technol*Shifting PM_{2.5} control from mass to toxicity for health equity worldwide*

Perspective synthesising toxicological, epidemiological and modelling evidence against the uniform-toxicity assumption underlying mass-based standards. The empirical hook is that age-standardised attributable death rates diverged worldwide over the past decade despite falling mass. Incomplete-combustion sources — residential solid fuel and wildfire above all — carry the highest toxicity per unit mass from primary emission through secondary ageing. Toxicity-adjusted exposure is substantially elevated in South Asia, Southeast Asia, sub-Saharan Africa, Andean Latin America and China, and **the 8.2% of the world that is low-income bears 24.0% of the cumulative toxicity-adjusted exposure burden**. It is a perspective, not a quantitative assessment: the toxicity weights are inherited from a heterogeneous literature and the equity arithmetic is only as good as they are.

doi: 10.1021/acs.est.6c05282

Lim et al. 2026 Nat Commun*Public knowledge, attitudes and practices on air quality in the ten most populous countries*

Quota sample of **10,618 respondents across 1,372 localities** in Bangladesh, Brazil, China, Indonesia, India, Mexico, Nigeria, Pakistan, Russia and the United States, linked to modelled PM_{2.5}. Exposure is *inversely* associated with knowledge about air quality but *positively* associated with protective practices, attitudes and concern — people in dirtier air act more and know less. Interactions with age, sex, education, residence, political ideology, marital, employment and smoking status are all significant. Quota sampling is not probability sampling, so national point estimates should not be read as prevalence; the cross-country contrasts are the durable part. A useful anchor for anyone arguing that a dense sensor network has a communication function as well as a measurement one.

doi: 10.1038/s41467-026-75908-7

Cardiovascular & metabolic

1 record

Wang L et al. 2026 Lipids

Joint exposure to small dense LDL-C and long-term PM_{2.5} and incident stroke

CHARLS 2011–2020, 6,457 stroke-free adults, 710 incident strokes over a median 9 y; PM_{2.5} from the ChinaHighPM_{2.5} product; Cox models with *k*-means trajectory clustering and restricted cubic splines. Per 1 SD: **sd-LDL-C HR 1.116 (1.018–1.224)**, **PM_{2.5} HR 1.321 (1.222–1.428)**, with a significant multiplicative interaction ($p=0.037$) and **HR 1.713 (1.335–2.198)** for the dual-high stratum against dual-low. A sustained high sd-LDL-C trajectory added 36% (1.361, 1.086–1.706). sd-LDL-C outperformed conventional LDL-C for discrimination (AUC 0.578 vs. 0.534, $p < 0.001$) — though both are poor in absolute terms, which the abstract does not say. sd-LDL-C is *estimated*, not measured, and PM_{2.5} is a modelled annual mean at residence, so the interaction is between two proxies and is the result most likely to attenuate under better measurement.

doi: 10.1002/lipd.70072

Other clinical endpoints

1 record

Feng et al. 2026 Joint Bone Spine

Air pollutant mixtures, plasma metabolic and proteomic signatures, and incident arthritis

UK Biobank, 401,676 arthritis-free at baseline, 78,188 incident cases over a median 13 y. Mixture exposure quantified by weighted quantile sum regression; metabolic and proteomic signatures derived by elastic net; Cox models plus generalised propensity scores; causal mediation for the signatures. The dominant mixture contributors are subtype-specific — NO_x/NO₂ for rheumatoid arthritis, PM for osteoarthritis, NO_x and PM for gout, PM for psoriatic arthritis — and the omics signatures mediated 4.92–35.05% of the association. The abstract reports no hazard ratios or intervals for the WQS index itself, which is why nothing from this paper appears in the forest plot. WQS indices are also not comparable across studies, and the mediation proportions rest on a no-unmeasured-confounding assumption that plasma omics cannot support. The subtype-resolved contributor pattern is the transferable result.

doi: 10.1016/j.jbspin.2026.106129

Mechanistic toxicology

1 record

García Dominguez et al. 2026 Environ Sci Technol

Ultrafine particle deposition and clearance in an avian lung

Zebra finches exposed to 50 or 100 nm model particles at 5 or 25 °C, all within environmentally relevant ranges, with deposition quantified in lung and — in a subset — heart, liver and red blood cells. Lung deposition was highest for the smallest particles, consistent with diffusional deposition. Deposited *fraction* was unchanged by temperature but deposited *dose rate* rose in the cold, driven entirely by elevated minute ventilation: a clean demonstration that thermoregulatory demand, not respiratory physiology, sets avian dose. **44% of the initially deposited 100 nm burden remained at two weeks**, i.e. clearance is slow. Model particles, one species, small subset for extrapulmonary organs — but this is the only record in today's corpus that measures ultrafine dose in a living respiratory system, and the avian lung's cross-current architecture makes it a genuinely different test case from rodent work.

doi: 10.1021/acs.est.6c05957

Corpus register

First author	Focus	Journal	Design	Metric	Region
Cardiovascular & metabolic					
Wang L et al.	Joint exposure to small dense LDL-C and long-term PM _{2.5} and risk of incident stroke	<i>Lipids</i>	Prospective cohort + trajectory clustering	Annual mean PM _{2.5} from the ChinaHighPM _{2.5} dataset, 2011-2020	China
Mechanistic toxicology					
Garcia Dominguez et al.	Ultrafine airborne particle deposition and clearance in an avian lung under ecologically relevant conditions	<i>Environ Sci Technol</i>	Animal inhalation exposure + dosimetry	50 and 100 nm model ultrafine particles at 5 or 25 degrees C	Sweden
Occupational & indoor					
Illa et al.	How are we exposed to chemicals when drugs are smoked indoors?	<i>Environ Sci Technol</i>	Mechanistic indoor fate and exposure model	Indoor smoke aerosol; 7 compounds spanning volatility and hydrophobicity; first-, second- and third-hand pathways	Global (computational)
Jin et al.	Synergistic optimization of air change rate and air purifier type for bioaerosol control during dental ultrasonic scaling	<i>Build Environ</i>	Metadata only	Bioaerosol transport and deposition during dental ultrasonic scaling	China
Wood et al.	Influence of ventilation rate on the potential for indoor particulate matter exposure	<i>Indoor Environments</i>	Metadata only	Indoor PM exposure potential as a function of ventilation rate	United Kingdom
Sensing, forecasting & instrumentation					
Liu T et al.	Aerosol solute strength drives transition-metal-ion-catalyzed sulfate aerosol formation	<i>Environ Sci Technol</i>	Aerosol flow reactor + model implementation	Aqueous aerosol sulfate from SO ₂ + O ₂ with Fe(III)/Mn(II) catalysis; ionic strength threshold 16 mol/kg	Global / multi-region
Saad et al.	Short-term monitoring of air pollutants at the port-urban exposure zone, Alexandria Port	<i>Environ Monit Assess</i>	Portable sensor monitoring campaign	PM _{2.5} , PM ₁₀ , CO ₂ , CO, NO ₂ , SO ₂ , NH ₃ and VOC by portable multi-gas monitor and particulate sensors	Egypt
Yohanna & Lim	Regional and seasonal variation of PM _{2.5} and ozone across Malaysia using XGBoost and SHAP	<i>Chemosphere</i>	Gradient-boosting model + SHAP interpretability	Daily PM _{2.5} and O ₃ from the DOE monitoring network plus ERA5 reanalysis, 2018-2023	Malaysia
Exposure assessment & modelling					
Boogaard et al.	Advancing long-term outdoor air pollution exposure assessment for health studies	<i>Curr Environ Health Rep</i>	Programme-level synthesis of exposure-model comparisons	Long-term outdoor PM _{2.5} , NO ₂ and BC; mobility- and infiltration-adjusted exposure models	Global / multi-region
Chi et al.	Heterogeneous nitrosation of amines driven by dinitrogen tetroxide: a missing source of particulate nitrosamines	<i>Atmos Chem Phys</i>	Born-Oppenheimer molecular dynamics + metadynamics	Particulate nitrosamines from methylamine / dimethylamine + N ₂ O ₄ at the air-water interface	Global (computational)
Guo et al.	Resolving systematic errors in sulfate source apportionment: a field-validated kinetic isotope fractionation framework	<i>Atmos Chem Phys</i>	Kinetic model + field validation	Seasonal PM _{2.5} sulfate; stable-isotope Bayesian mixing plus Rayleigh fractionation	China
Hernandez Pardo et al.	Downward transport of tropical upper-tropospheric aerosols: multi-year idealized simulations	<i>Atmos Chem Phys</i>	Global chemistry-climate tracer simulation	19 idealised aerosol tracers, 20-100 nm, released in the tropical upper troposphere	Global / multi-region

First author	Focus	Journal	Design	Metric	Region
Li Q et al.	Health risk of fine particulate matter across China with a pollutant-specific index, 2013-2023	<i>Environ Sci Technol</i>	Machine-learning exposure model + meta-analysis + risk index	Daily 1 km PM _{2.5} , 2013-2023, machine-learning estimate; PM _{2.5} -specific AQHI	China
Burden, policy & mitigation					
Lim et al.	Public responses about air quality in the world's ten most populous countries	<i>Nat Commun</i>	Multi-country quota survey + exposure linkage	Modelled ambient PM _{2.5} linked to respondent locality	Global / multi-region
Song et al.	Shifting PM _{2.5} control from mass to toxicity for health equity worldwide	<i>Environ Sci Technol</i>	Perspective / evidence synthesis	Source-specific PM _{2.5} toxicity weighting applied to global exposure surfaces	Global / multi-region
Other clinical endpoints					
Feng et al.	Plasma metabolic and proteomic signatures of mixed ambient air pollutant exposure and incident arthritis subtypes	<i>Joint Bone Spine</i>	Prospective cohort (mixtures) + omics mediation	Annual PM _{2.5} , PM ₁₀ , NOx, NO ₂ combined in a weighted quantile sum index	United Kingdom

Provenance and method

Window: 31 August 2026 alone, contiguous with the 30 August issue; no gap. Sources queried on 1 September 2026, before 22:00 local, so 31 August is the newest buildable date under the run-timing rule. **PubMed** [EDAT] 2026/08/31 — health axis **11** via the connector; the instrumentation axis had to be split in two because the connector caps boolean operators at 20, and both surviving hits (42674653, 42673782) were already in the health set, so the sensing axis added **0** unique records. **Europe PMC** CREATION_DATE 2026-08-31 — 1 hit (PMID 42669197, World Trade Center dust), **already shipped on 30 August**; both the pmid and doi legs of `seen.json` caught it. **Crossref by-ISSN** across the tracked non-PubMed journals — **35** deposits with a 31 August created date, of which 8 are carried and 27 screened out (3 retraction notices, the rest PFAS, nanoplastics, battery recycling, MOF catalysis, GNSS meteorology and gas-phase ozone work). **arXiv** relevance query returned 60 entries, **none** published on or after 26 August. **OpenAlex remains dead** — HTTP 403 paid-plan block — and is routed through **Consensus**, which returned 3 hits on the calibration/drift axis: **all three** resolved to DOIs already in `seen.json` (10.1088/1361-6501/ae6298, 29 Jul; 10.3390/atmos17040335, 2 Aug; 10.5194/amt-19-603-2026, 1 Aug), so Consensus contributed no backfill this run. **47 unique records considered; 1 dropped as already seen; 16 in scope; 31 logged to rejected.json1 with a reason.** *Two records are listed on metadata only* — 10.1016/j.buildenv.2026.115193 and 10.1016/j.indenv.2026.100191 deposited no abstract to Crossref and are absent from PubMed, Europe PMC and Semantic Scholar; they are not summarised and no numbers are attributed to them. Subtopic, design, particle-metric and geography labels are assigned from title, abstract, keywords and MeSH terms; assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. First-author surnames are read from PubMed `efetch` or the Crossref deposit, never inferred. **Trial watch**, window 31 August 2026, ClinicalTrials.gov API v2 on `LastUpdatePostDate`: 1 hit, 1 kept — NCT07797413 (Savannah Environmental Nutrition Study, a food-based antioxidant intervention near the Port of Savannah, *n*=20, not yet recruiting, feasibility endpoints with FeNO and exhaled-breath-condensate H₂O₂ as secondary markers). `analyze_endpoints` was run; the record is surfaced in the monthly rollup, not here. Content derived from PubMed and Crossref metadata; abstracts remain the copyright of their respective publishers.